Classical action scales: obstructions, conditional bounds and quantum premises

Classical mechanics in the tested classes permits action-valued observables arbitrarily close to zero. Positive bounds appear when the class supplies an excitation floor, a fluctuating reference preparation, or restricted measurement information. None of the accepted calculations identifies such a bound with a universal quantum phase parameter. The useful outcome is a map of the premises that a selection principle must add and the counterexamples it must exclude.

This paper consolidates accepted repository results through C123. Its claims retain the domains of their original proofs; it states no combined theorem covering every classical model. Proof and literature status remain in the claim ledger. The final section gives an evidence map for the inherited audits. The detailed receiver/calibration sequence is a technical resource for the questions summarized here.

1. Select the observable before asking whether it has a floor

An action difference, orbital action, fluctuation coefficient and reconstruction error product can all have action units while answering different questions. We use physical units throughout, with Planck's constants related by $h = 2\pi\hbar$; a model parameter K is not identified with either by notation.

Object	Definition and domain	What positivity would mean
Action excess	$\Delta S = S[q+\eta] - S[q]$, on a stated fixed-endpoint path class	Separation in action values on that class
Orbit action	$J = (2\pi)^{-1} \oint p \cdot dq$, on closed trajectories	Minimum integrated mechanical excursion
Displacement response	$h_\Delta = m \operatorname{Var}[X(t+\Delta) - X(t)]/\Delta$	Prepared fluctuations at observation window Δ
Canonical reconstruction product	$R_x R_P$, optimal worst-case scalar errors for stated data and preparations	Limits of that information protocol
Spectral gap	Lowest positive spectrum of a specified operator	Separation from its ground sector; energy or inverse-time units depend on the operator

For a stationary confined coordinate of finite variance, the displacement response tends to zero as the observation window grows, even when its orbit action is positive. C047 and C054–C055 make this observable mismatch concrete. For reconstruction, positive coordinate errors can occur on a smooth compatible curve of zero projected planar area (C120). An error product is therefore not a phase-space cell area without an additional argument.

2. Three routes to arbitrarily small classical action

Continuous variation closes action excess. With mass $m > 0$, duration $T > 0$, constant force and fixed endpoints, the variation $\eta(t) = at(T-t)$ gives

$$\Delta S = \frac{ma^2T^3}{6} \longrightarrow 0 \quad (a \longrightarrow 0).$$

The varied paths are comparison paths, not generally solutions of the same boundary-value dynamics. This is C002's precise admissibility boundary. A positive Dirichlet Hessian controls cost relative to the variation norm; the variation amplitude can still shrink. The foundations paper contains the calculation and its source audit.

Admitted contractions close several action observables together. For Newtonian or relativistic kinetic energy $T(p)$ and external potential $V(q)$, set

$$V_a(Q) = V(Q/a), \qquad q_a(t) = aq(t/a), \qquad p_a(t) = p(t/a), \quad a > 0.$$

Hamilton's equations are preserved between the two models. Mass, speeds and energy values stay fixed; forces scale by a^{-1}. Canonical orbit action and Hamilton action scale by a. Corresponding prepared covariance actions and corresponding observation-window responses obey the same degree-one scaling. Thus any admissible class closed under arbitrarily small such contractions has zero positive-action infimum (C056–C057). A fixed potential or uniform force ceiling can exclude this particular transformation. See the dilation proof for the full domain and limit order.

Even a fixed force-bounded potential can admit small actions. C058 fixes one smooth relativistic confining potential

$$V(r) = k_s b^2 \left(\sqrt{1 + r^2/b^2} - 1 \right), \qquad k_s, b, m, c > 0.$$

Here k_s is stiffness; the force is globally bounded by $k_s b$. The stable circular orbits have $J_R \sim \sqrt{mk_s}R^2 \to 0$ as $R \to 0$; their speeds also tend to zero. This family keeps the potential fixed, so excluding the contraction map alone does not supply a floor. The fixed-force proof isolates the missing excitation premise.

These are complementary countertests with different admissibility assumptions. A proposed principle should state which family it excludes and which independent physical condition excludes it.

3. Positive classical bounds reveal their physical input

Persistent excitation plus limited force gives a sharp orbit-action bound. For a regular closed C^2 trajectory in dimension at least two, canonical momentum $p = m\dot{q}$, speed at least $v_* > 0$ and force at most $F_{\max} > 0$, C060 gives

$$J \geq \frac{m^2 v_*^3}{F_{\max}}.$$

Total turning is at least 2π; the force ceiling limits the rate of momentum turning, and the speed floor converts that duration into action. A circle realizes equality within the admissible smooth-potential class. For relativistic momentum the bound is multiplied by $(1 - v_*^2/c^2)^{-1}$, with $v_* < c$. The excitation floor supplies positivity; the scale depends on mass, excitation and force. The closed-orbit proof retains the momentum, frame and regularity assumptions. A19's peak-excursion variant is a supporting question, not yet an accepted result.

Composition propagates a supplied positive fluctuation scale. For independent constituents and a mass-only coefficient within one preparation class, centre-of-mass composition gives

$$\kappa(m_1 + m_2) = \frac{m_1\kappa(m_1) + m_2\kappa(m_2)}{m_1 + m_2}.$$

Nonnegativity and additive composition for every positive mass force $\kappa(m) = K \geq 0$. A reference two-state velocity process with mass m_0, speed $u_0 > 0$ and finite positive reversal rate λ_0 then gives $K = m_0 u_0^2/\lambda_0 > 0$ (C035–C036). The reference preparation supplies positivity. Different preparation classes can have different K; matching a whole-body preparation to independently composed constituents is an explicit premise. The composition proof also gives countertests of that premise.

The singular relativistic Kepler threshold $|L| > k/c$ is another useful positive classical benchmark, inherited from established literature. Its value depends on the supplied coupling and its regular collision-free domain. Softening the core admits circles of arbitrarily small angular action (C052–C053). It does not supply an action lattice or a universal constant. See the Kepler comparison.

4. What the calibration sequence settles

Preparation and record access decide the reconstruction obstruction in the finite mechanical receiver. R05–R33 make those resources explicit; their principal contribution is the following classification.

Information/preparation regime	Accepted conclusion	Selection premise still needed
Scalable incoming preparation and exact delayed records	A fixed finite-mass, finite-duration apparatus allows vanishing reconstruction error and disturbance, C068–C069	Physical restriction preventing that preparation/record limit
Fixed preparation uncertainty with incomplete supplied data	Exact common-record families give positive reconstruction risks in specified classes, C090–C093 and C106–C115	Reason those uncertainties are unavoidable and universal
Sufficient exact calibration and full records on the stated fixed box	Uniform receiver recovery, with error bounded by record error divided by coupling, C116–C118	An independent restriction on final-record or calibration precision
Four finite calibration tolerances with exact records	Canonical product has matched order $\min\{1, (\epsilon/\lambda)^2\}$ in fixed component units, C119	Origin and composition law of tolerance epsilon
One uncertain or correlated calibration direction	The direction determines canonical response; a zero-area curve can have positive coordinate risks, C120–C123	A physical reason for the admitted uncertainty geometry

The recovery row uses R30's sufficiently small fixed apparatus box and bounded convex receiver domain. It does not cover R29's larger preparation box. The R33 estimates concern a selected local common-record fibre; their upper bounds are not global minimax bounds. All error products convert to action through the fixed physical position and momentum units.

C123 adds an exact symmetry cancellation and explicit quadratic response. Its remaining curvature coefficient may refine a local error law, but no current argument makes that coefficient decide quantum necessity or universality. The research decision is to park that continuation and retain its derivation for a named future dependency. The R33 handoff preserves the open calculation.

The general lesson is conditional: these apparatus examples defeat a derivation that relies only on the resource bounds they satisfy. Positive classical reconstruction risk and the quantum commutation relation remain distinct claims. The apparatus construction and calibration-tolerance proof supply the underlying dynamics and estimates.

5. Quantum necessity and mass gap require different next arguments

Quantum necessity requires a premise that excludes the classical alternatives. The checkerboard comparison starts with complex amplitudes, Born probabilities and a supplied $K > 0$, then derives its Dirac continuum limit. It establishes what those premises produce; it does not select them mechanically. Q01 now asks which reconstruction premise excludes a classical state space, while K01 tests how fixed-time measurement compatibility behaves when apparatus variables are included. The later dynamical step must still locate action units, positivity and universality. The coherent-path comparison and B01 source coverage define the starting point; the reconstruction proofs have not yet received a full repository audit.

The spectral route requires control of all slow modes. For a finite irreducible reversible generator, put $A = -Q$ on its centered Hilbert space and $\chi(f) = \langle f, A^{-1} f \rangle$. If an observable collection satisfies

$$\sum_a |\langle f_a, g \rangle|^2 \geq \alpha \|g\|^2, \qquad S = \sum_a \chi(f_a) < \infty, \qquad \alpha > 0,$$

then its relaxation gap obeys $\gamma \geq \alpha / S$ (C042). For velocity-unit observables, alpha has speed-squared units, S has length-squared/time units, and gamma has inverse-time units. The proof applies the coverage inequality to a slowest eigenvector. A single positive action plateau can miss that vector; C041 and C045 exhibit closing gaps with a fixed plateau. Full rank at each parameter is insufficient when calibration becomes arbitrarily weak. Independent product dynamics control mixed modes (C046).

G03 asks for an interaction estimate preserving a gap uniformly as system size increases. The observable coverage and response constants must also be controlled; a finite-dimensional trace bound can deteriorate with size. The susceptibility proof records the exact hypotheses.

A Yang–Mills application additionally needs the physical Hamiltonian and quantum theory, gauge-invariant observables, and continuum and infinite-volume control. An auxiliary sampling clock can change a relaxation gap without changing the invariant law. The field-theory comparison keeps these operator and time identifications explicit. Neither a positive classical action observable nor a toy relaxation gap supplies that construction.

6. Evidence map and next decisions

Consolidated argument	Claims and written proof	Inherited source/proof audits
Vanishing variation and mechanical contraction	C002, C056–C058; section 2 links	B05, B30, B31
Conditional excitation, composition and Kepler bounds	C035–C036, C052–C053, C060–C061; section 3 links	B16, B27–B28, B32
Preparation and records	C068–C069, C090–C123; section 4 links and ledger	B37, B48–B66
Coherent versus probabilistic path rules	C039–C040; checkerboard note	B18
Slow modes, calibrated coverage and product control	C041–C042, C045–C046; susceptibility note	B20, B22

Q01's premise map identifies reversible pure-state connectivity and purification as distinct classical-exclusion premises (B67). Hardy's moving-ball example motivates the next quantum question: exact operational closure during transformations, with action identification still separate. G03's interacting size-uniform gap test is selected next in STATE.